DSP – LEC – 7 - Ch-3

CHAPTER - 3

CRRELATION AND CONVOLUTION

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CORRELATION AND CONVOLUTION

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Sec. 3.1

3.1 CORRELATION DESCRIPTION

- Consider how two data sequences, each consisting of simultaneously sampled values taken from the two corresponding waveforms, might be compared.

- If the two waveforms varied similarly point for point, then a measure of their correlation might be obtained by taking the sum of the products of the corresponding pairs of points.

$$r_{12} = \sum_{n=0}^{N-1} x_1(n)x_2(n) \qquad (3.1)$$

Sec. 3.1 Cont. - 1

- This proposal becomes more convincing when the case of two **independent and random** data sequences is considered.

- This sum of the products will tend towards a small random number as the number of pairs of points is increased.

- This is because all numbers, positive and negative, are equally likely to occur so that the product pairs tend to be self-cancelling on summation.

Sec. 3.1 Cont. - 2

- By contrast, the existence of a finite sum will indicate a degree of correlation.

- A negative sum will indicate negative correlation, which is an increase in one variable is associated with a decrease in the other variable.

- The cross-correlation $r_{12}(n)$ between two data sequences $x_1(n)$ and $x_2(n)$ each containing N data might therefore be written as

$$r_{12} = \sum_{n=0}^{N-1} x_1(n)x_2(n) \qquad (3.1)$$

Sec. 3.1 Cont. - 3

- This definition, produces a result which depends on the number of sampling points taken.

- This is corrected for by normalizing the result to the number of points by dividing by N.

- Thus, an improved definition is

$$r_{12} = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) x_2(n) \qquad (3.2)$$

Sec. 3.1 Cont. - 4

Example 3.1

Calculate r_{12} of x_1 and x_2, in which the point numbers in the data sequences are "n".

n	1	2	3	4	5	6	7	8	9
x_1	4	2	-1	3	-2	-6	-5	4	5
x_2	-4	1	3	7	4	-2	-8	-2	1

$$r_{12} = \frac{1}{9}(4\times -4 + 2\times 1 + -1\times 3 + 3\times 7 + -2\times 4 + -6\times -2 + -5\times -8 + 4\times -2 + 5\times 1) = 5$$

Sec. 3.1 Cont. - 5

- In some cases it may indicate zero correlation although the two waveforms are 100% correlated.

- This may occur for example when the two waveforms are out of phase, which will often be the case. The situation is illustrated by the waveforms of **Fig. 3.1**.

Sec. 3.1 Cont. - 6

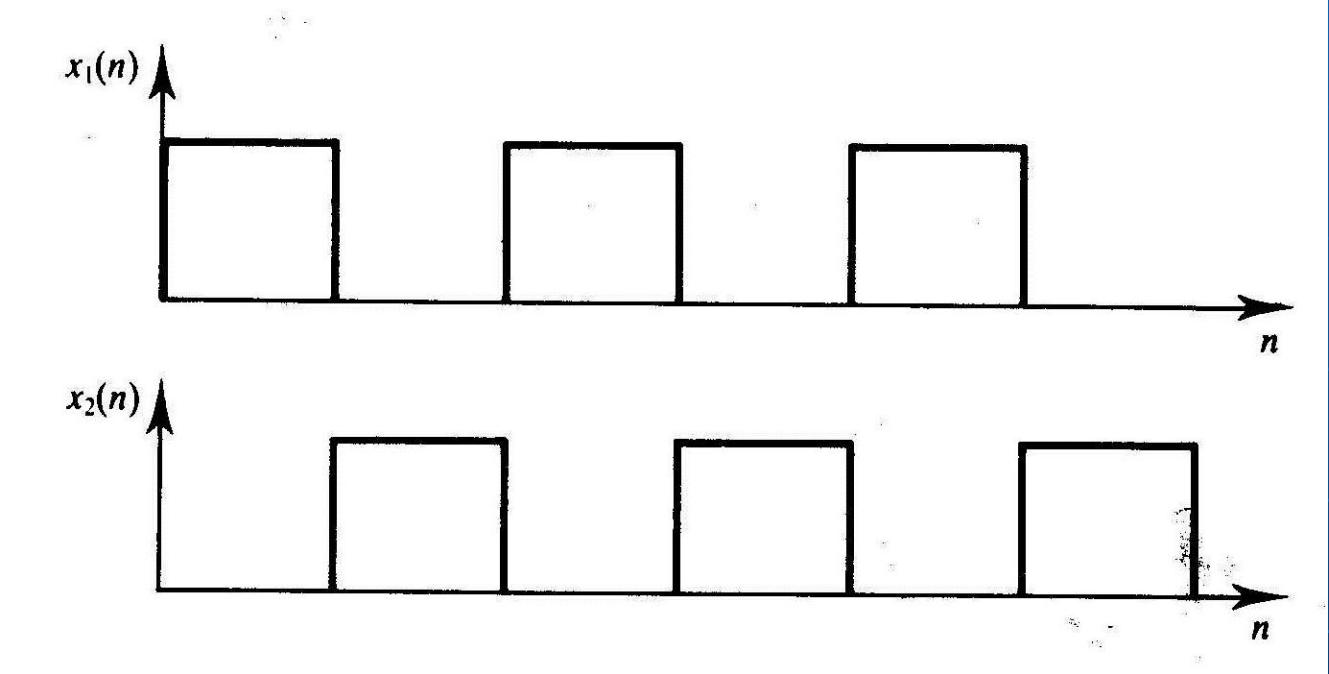


Fig. 3.1 Out-of-phase 100% correlated waveforms with zero correlation at lag zero

Sec. 3.1 Cont. - 7

- From this figure it is seen that each pair product in the correlation is zero, and hence the correlation is zero, because one of either x_1 or x_2 is always zero.

- However, the waveforms are clearly highly correlated, although they are out of phase.

- To overcome such phase differences it is necessary to shift, or lag, one of the waveforms with respect to the other.

Sec. 3.1 Cont. - 8

- As illustrated in **Fig. 3.2** this is equivalent to changing $x_2(n)$ to $x_2(n+j)$, where j represents the amount of lag which is the number of sampling points by which x_2 has been shifted to the left.

- An alternative, but equivalent, procedure is to shift x1 to the right.

- The formula for the cross correlation thus becomes:

$$r_{12}(j) = \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)x_2(n+j)$$

$$= r_{21}(-j) = \frac{1}{N}\sum_{n=0}^{N-1} x_2(n)x_1(n-j) \qquad (3.3)$$

Sec. 3.1 Cont. - 9

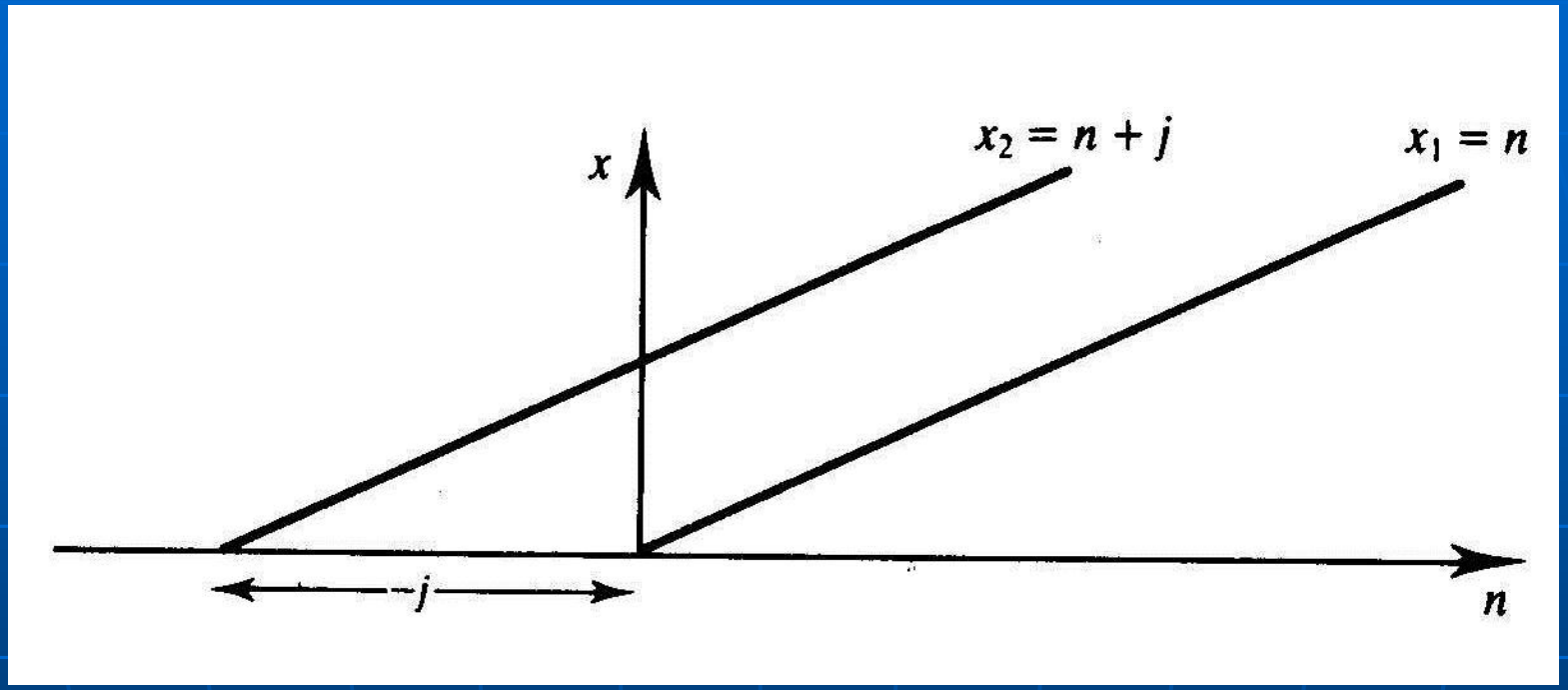


Fig. 3.2 Waveform $x_2 = x_1 + j$ shifted j lags to the left of waveform x_1.

Sec. 3.1 Cont. - 10

Example 3.2

Consider the cross-correlation of the above two sequences $x_1(n)$ and $x_2(n)$ at a lag of j = 3 , that is consider $r_{12}(3)$. The two sequences become

n	1	2	3	4	5	6	7	8	9
x_1	4	2	-1	3	-2	-6	-5	4	5
x_2	7	4	-2	-8	-2	1			

So

$$r_{12}(3) = \frac{1}{9}(4\times 7 + 2\times 4 + -1\times -2 + 3\times -8 + -2\times -2 + -6\times 1)$$

$$= 1.334$$

Sec. 3.1 Cont. - 11

- Another problem associated with cross-correlating finite lengths of data.

- This can be seen in the above example in which $r_{12}(3) = 1.334$ was determined.

- As x_2 is shifted to the left the waveforms no longer overlap and data at the ends of the sequences no longer form pair products.

- This is known as **the end effect**.

Sec. 3.1 Cont. - 12

- This is known as **the end effect problem**.

- In the example the number of pair has dropped from nine to six for a lag of three.

- The result is a linear decrease in $r_{12}(j)$ as j increases, leading to debatable values of $r_{12}(j)$.

- As shown in **Fig. 3.3**.

Sec. 3.1 Cont. - 13

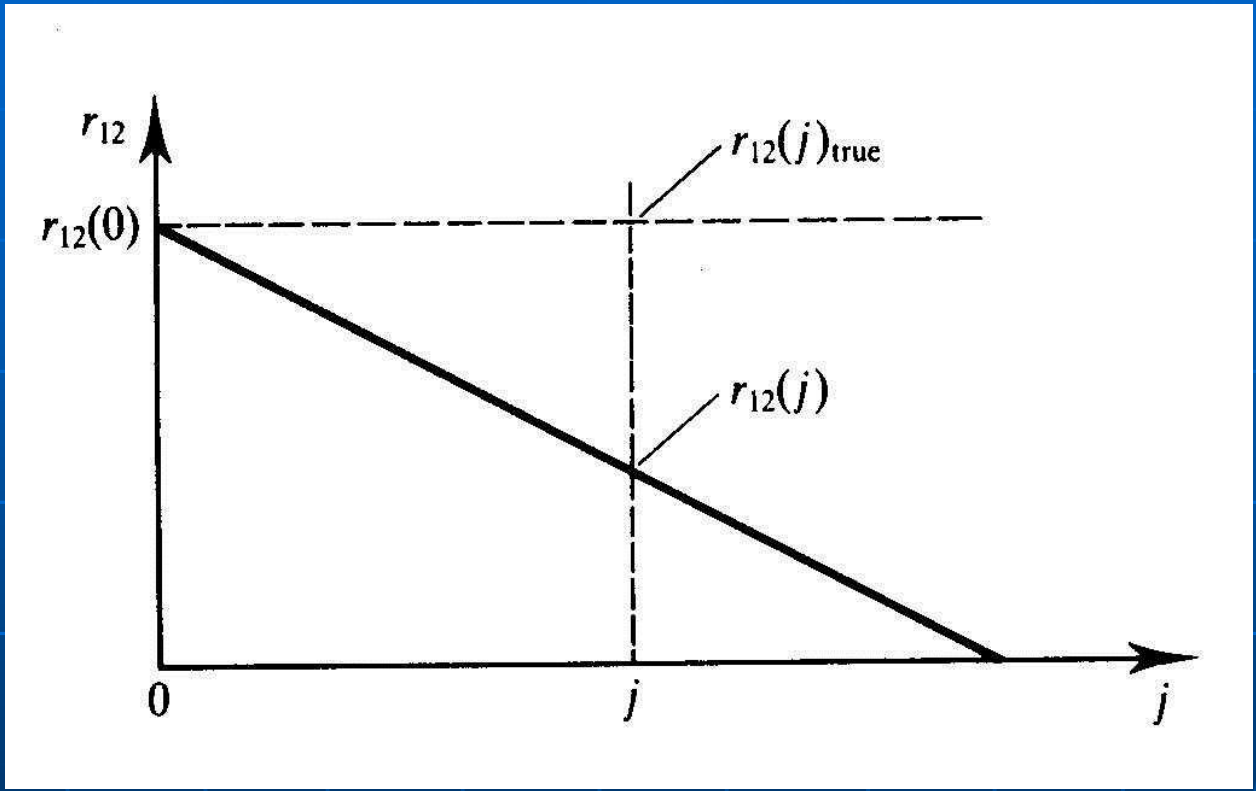


Fig. 3.3 The effect of the end-effect on the cross-correlation r12(j)

Sec. 3.1 Cont. - 14

- **First solution** is to make one of the sequences twice as long as the required length for correlation.

- This could be achieved by recording more data, or, if one of the sequences were periodic, by repeating the sequence (taking care to match the two ends).

- **Second solution** is to add a correction to all computed values.

- **Figure 3.3** shows how $r_{12}(j)$ decreases with j purely as a result of the end effect problem.

Sec. 3.1 Cont. - 15

- At **j=0**,

  $r_{12}(j) = r_{12}(0)$.

- At **j = N,**

  $r_{12}(N) = 0$,

  because the waveforms no longer overlap.

- In between, at some lag **j**, the true value of $r_{12}(j)$ is $r_{12}(j)$ true while the actual value caused by the end effect is $r_{12}(j)$.

Sec. 3.1 Cont. - 16

- Then, from Fig. 3.3, we find that:

$$\frac{r_{12}(j)_{true} - r_{12}(j)}{j} = \frac{r_{12}(0)}{N}$$

Where

$$r_{12}(j)_{true} = r_{12}(j) + \frac{j}{N} r_{12}(0) \qquad (3.4)$$

- Computed values of the cross-correlation are therefore easily corrected for end effects by adding $\frac{j}{N} r_{12}(0)$ to the values of $r_{12}(j)$ (actual value).

Correlation in continuous time domain:

- Of course, it is also possible to consider correlation in the continuous time domain, and some analogue signal correlation is implemented this way.

- In the continuous domain n→t and j→τ and

$$r_{12}(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x_1(t)x_2(t+\tau)dt \qquad (3.5)$$

Sec. 3.1 Cont. - 18

- However, if $x_1(t)$ and $x_2(t)$ are periodic with period T_0 Eq. 3.5 simplifies to

$$r_{12}(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_1(t) x_2(t+\tau)\, dt \qquad (3.6)$$

Sec. 3.1 Cont. - 19

- If the waveforms are of finite energy, e.g., non periodic pulse then, the average evaluated over the time T as $T \rightarrow \infty$ is not taken because when $1/T \rightarrow 0$ $r_{12}(\tau)$ is always vanishingly small.

- For this case Eq. 3.7 is used in principle :

$$r_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t)x_2(t+\tau)\,dt \qquad (3.7)$$

Sec. 3.1 Cont. - 20

- In practice, a finite record length will be processed and so **Eq. 3.8** or **Eq. 3.3** will be applied:

Correlation in continuous time domain, Eq. 3.8:

$$r_{12}(\tau) = \frac{1}{T}\int_0^T x_1(t)x_2(t+\tau)\,dt \qquad (3.8)$$

Correlation in discrete time domain, Eq. 3.3:

$$r_{12}(j) = \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)x_2(n+j)$$

$$= r_{21}(-j) = \frac{1}{N}\sum_{n=0}^{N-1} x_2(n)x_1(n-j) \qquad (3.3)$$

Sec. 3.1 Cont. - 21

Analytical cross-correlation:

- Cross-correlation may also be performed analytically when analytical expressions can be written for the waveforms **(using eq. 3.8)**.

$$r_{12}(\tau) = \frac{1}{T}\int_0^T x_1(t)x_2(t+\tau)\,dt \qquad (3.8)$$

- This is shown in **example (3.3)**.

EXAMPLE - 3.3

Example 3.3

Obtain the cross-correlation $r_{12}(-\tau)$ between the waveforms $v_1(t)$ and $v_2(t)$ of Fig. 3.6.

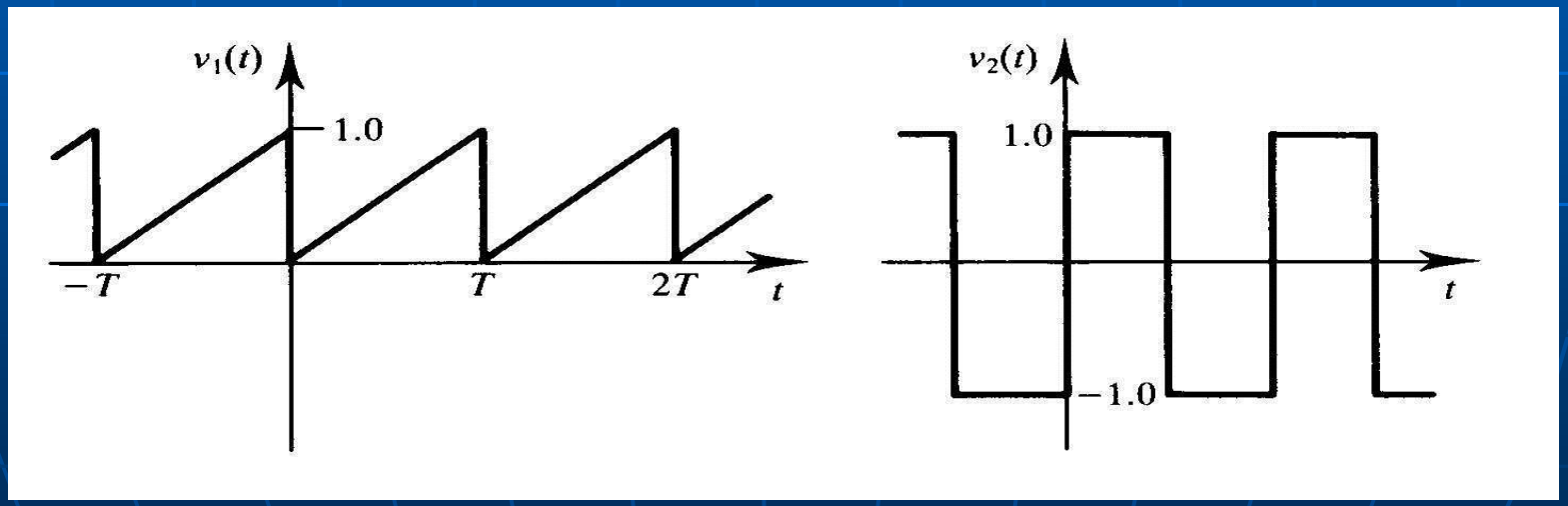


Fig. 3.6 The waveforms $v_1(t)$ and $v_2(t)$.

Example 3.3 - Cont - 1

Solution

- It is easy to express the waveforms analytically by dividing them into straight-line sections.

- It is only necessary to do this over one period, T, of the waveforms because r12(- τ) will be periodic in τ with period T.

Example 3.3 - Cont - 2

- Therefore,

for $0 \le t \le T$, $v_1(t)$ = t/T, and

for $0 \le t \le T/2$, $v_2(t)$ = 1.0, while

for $T/2 \le t \le T$, $v_2(t)$ = -1.0.

As shown in **Fig. 3.6**.

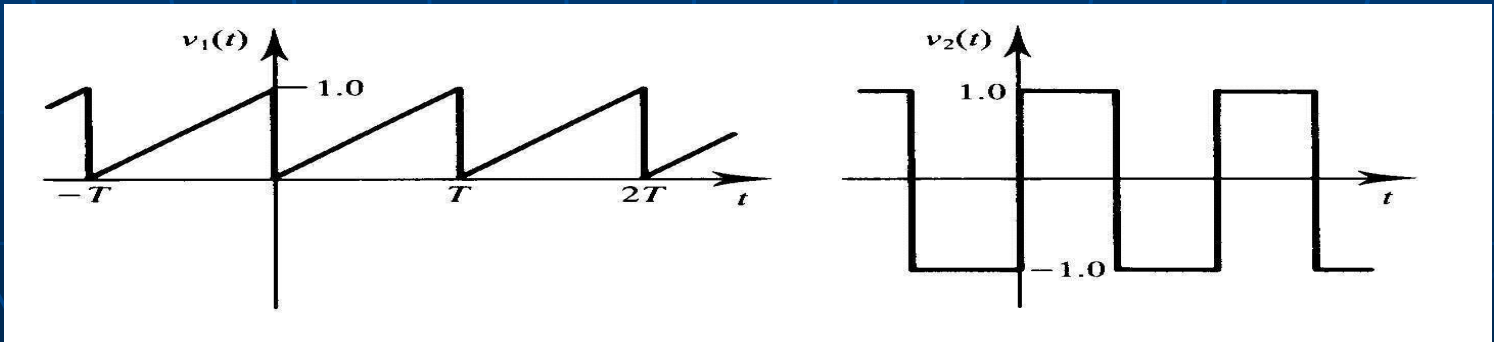
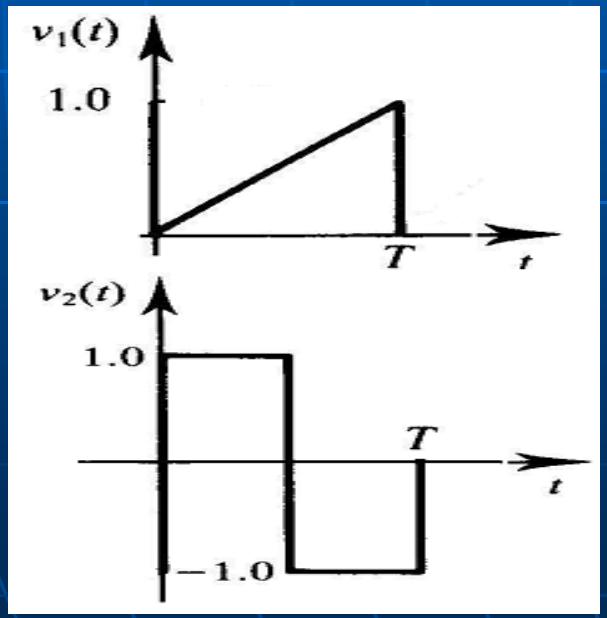


Fig. 3.6

Example 3.3 - Cont - 3

- It is required to obtain an expression for r12 (- τ), that is $v_2(t)$, is to be shifted right w.r.t. $v_1(t)$.

For $\tau = 0$ Fig. A



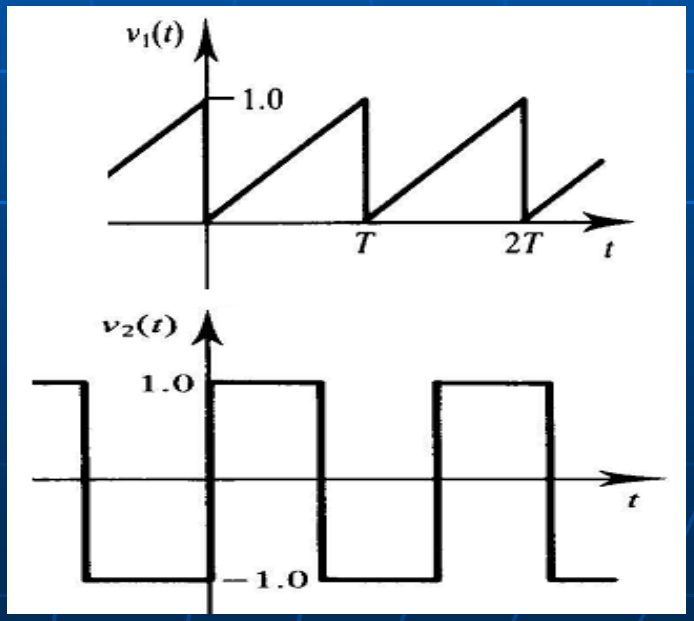


Fig. A

Example 3.3 - Cont - 4

For $0 \le \tau \le T/2$ Fig. 3.7

the situation shows that $\upsilon_1(t)$ has to be multiplied by three consecutive section of $\upsilon_2(t)$ in which it has the consecutive value (–1, 1, -1).

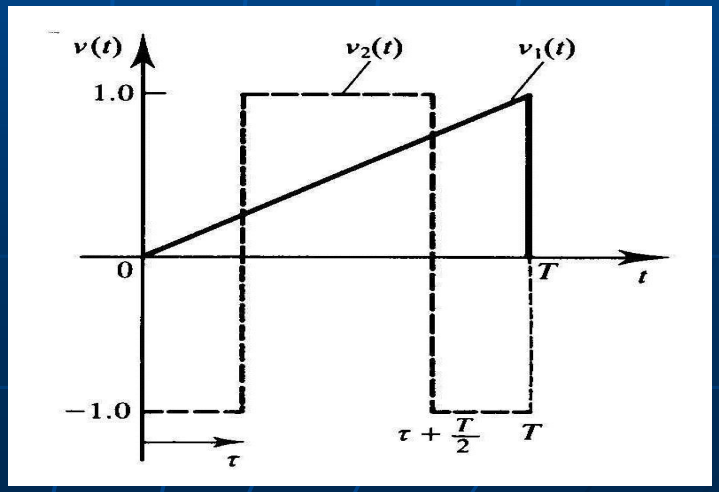


Fig. 3.7 Section of $\upsilon_2(t)$ for $0 \le \tau \le T/2$

Example 3.3 - Cont - 5

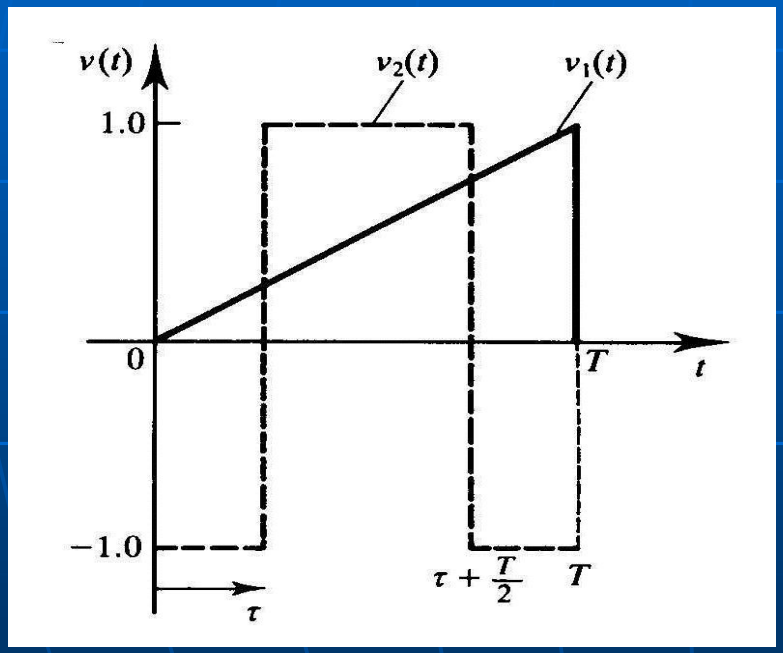


Fig. 3.7 Section of $v_2(t)$ for $0 \le \tau \le T/2$

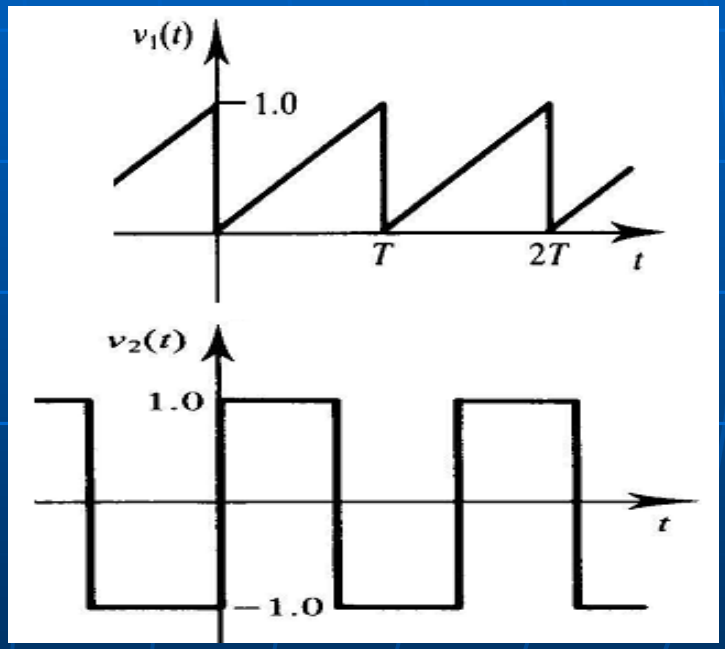


Fig. A

Example 3.3 - Cont - 6

- Referring to **Fig. 3.7**, the cross-correlation is split into the three sections with boundaries at

t = τ , t = τ +T/2, **and** t = T.

- Hence

$$r_{12}(-\tau) = \frac{1}{T}\int_0^T v_1(t)v_2(t-\tau)\,dt$$

$$= \frac{1}{T}\int_0^{\tau}\frac{t}{T}(-1)\,dt + \frac{1}{T}\int_{\tau}^{\tau+T/2}\frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau+T/2}^{T}\frac{t}{T}(-1)dt$$

$$= \frac{-1}{T^2}\left[\frac{t^2}{2}\right]_0^{\tau} + \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau}^{\tau+T/2} - \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau+T/2}^{T}$$

So,

$$r_{12}(-\tau) = -\frac{1}{4} + \frac{\tau}{T} \qquad \text{for} \qquad 0 \le \tau \le \frac{T}{2} \tag{3.9}$$

Example 3.3 - Cont - 7

For $T/2 \le \tau \le T$ Fig. 3.8

- The consecutive values of $v_2(t)$ have changed to 1, -1, +1.

- This means that, there are two parts to the solution which must match at τ = T/2.

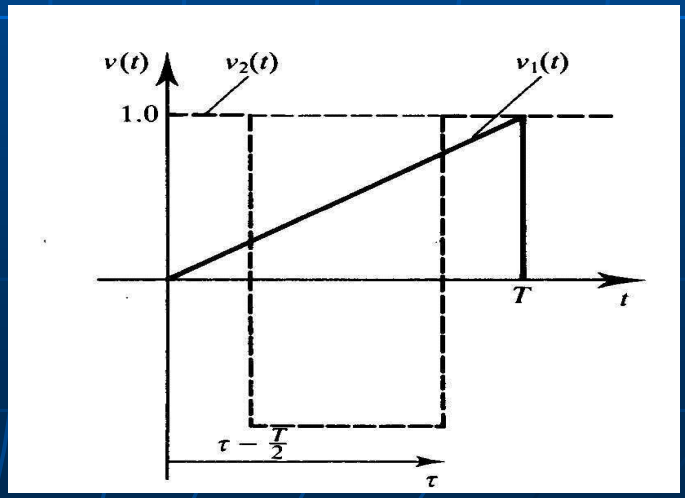


Fig. 3.8 Sections of $v_2(t)$ for $T/2 \le \tau \le T$.

Example 3.3 - Cont - 8

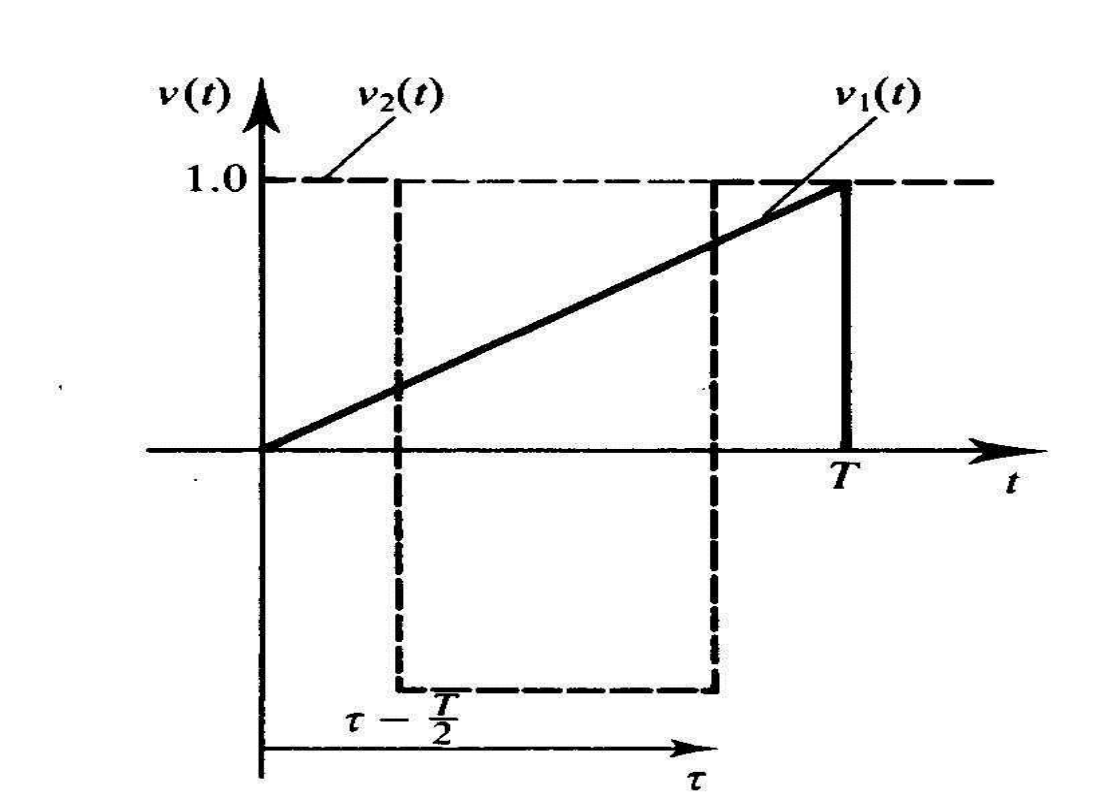


Fig. 3.8 Sections of $v_2(t)$ for $T/2 \le \tau \le T$.

Example 3.3 - Cont - 9

For $T/2 \le \tau \le T$ **and referring to Fig. 3.8 it is seen that:**

$$r_{12}(-\tau) = \frac{1}{T}\int_0^{\tau - T/2} \frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau - T/2}^{\tau} \frac{t}{T}(-1)dt + \frac{1}{T}\int_{\tau}^{T} \frac{t}{T}(1)dt$$

So,

$$r_{12}(-\tau) = \frac{3}{4} - \frac{\tau}{T} \qquad \text{for} \qquad \frac{T}{2} \le \tau \le T \qquad (3.10)$$

Example 3.3 - Cont - 10

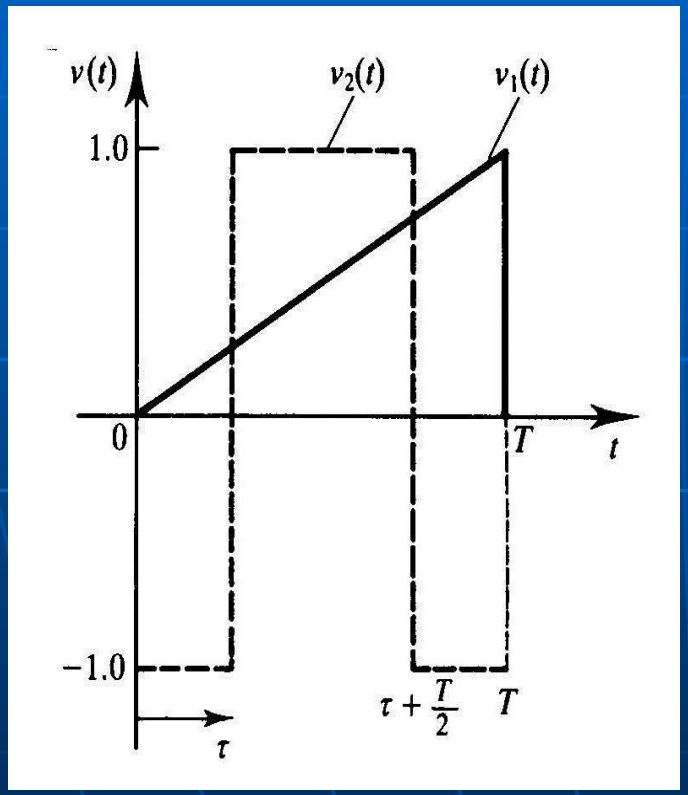


Fig. 3.7 Section of $v_1(t)$ for $0 \le \tau \le T$

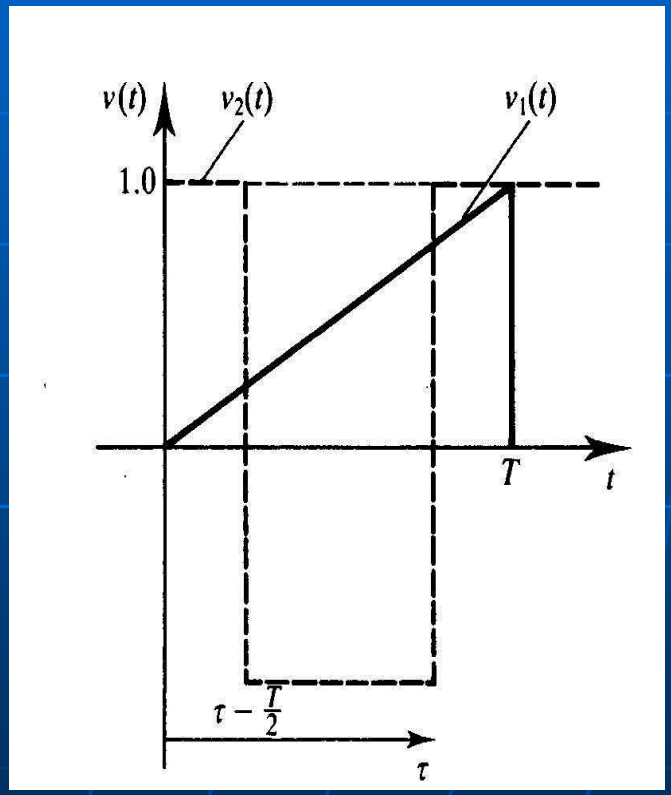


Fig. 3.8 Sections of $v_1(t)$ for $T/2 \le \tau \le T$

Example 3.3 - Cont - 11

- Substituting τ = T/2 into Eq.s 3-9 and 3-10 gives $r_{12}(-\tau) = 1/4$ in both cases, confirming that the two functions match correctly.

- **Figure 3.9** shows a plot of $r_{12}(-\tau)$ versus for $0 \le \tau \le T$.

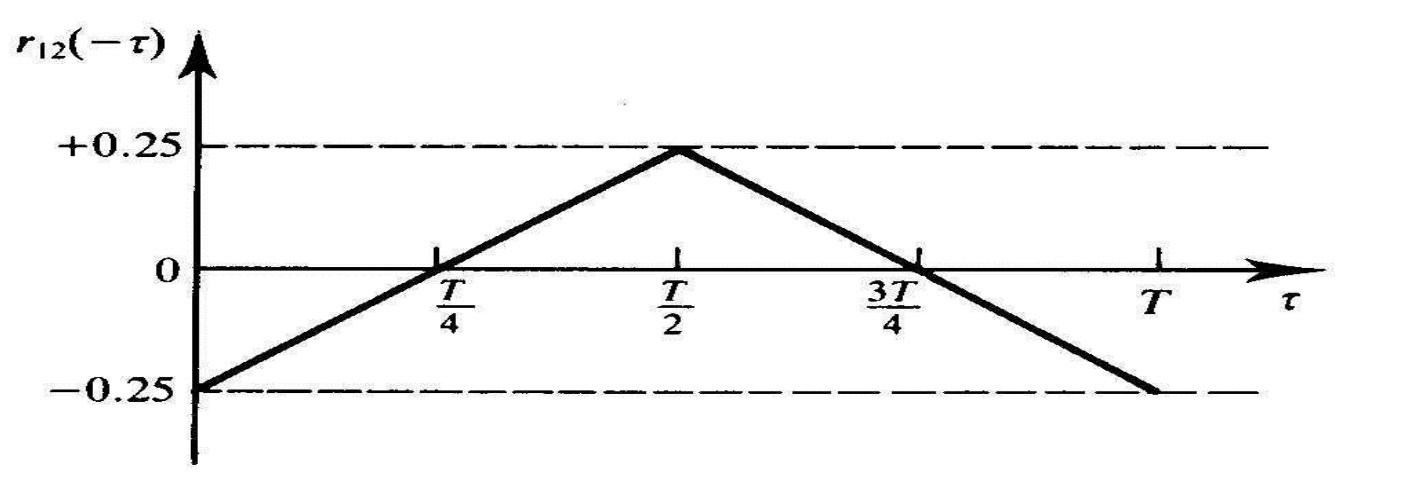


Fig. 3.9 $r_{12}(-\tau)$ as a function of τ .

Example 3.3 - Cont – 12

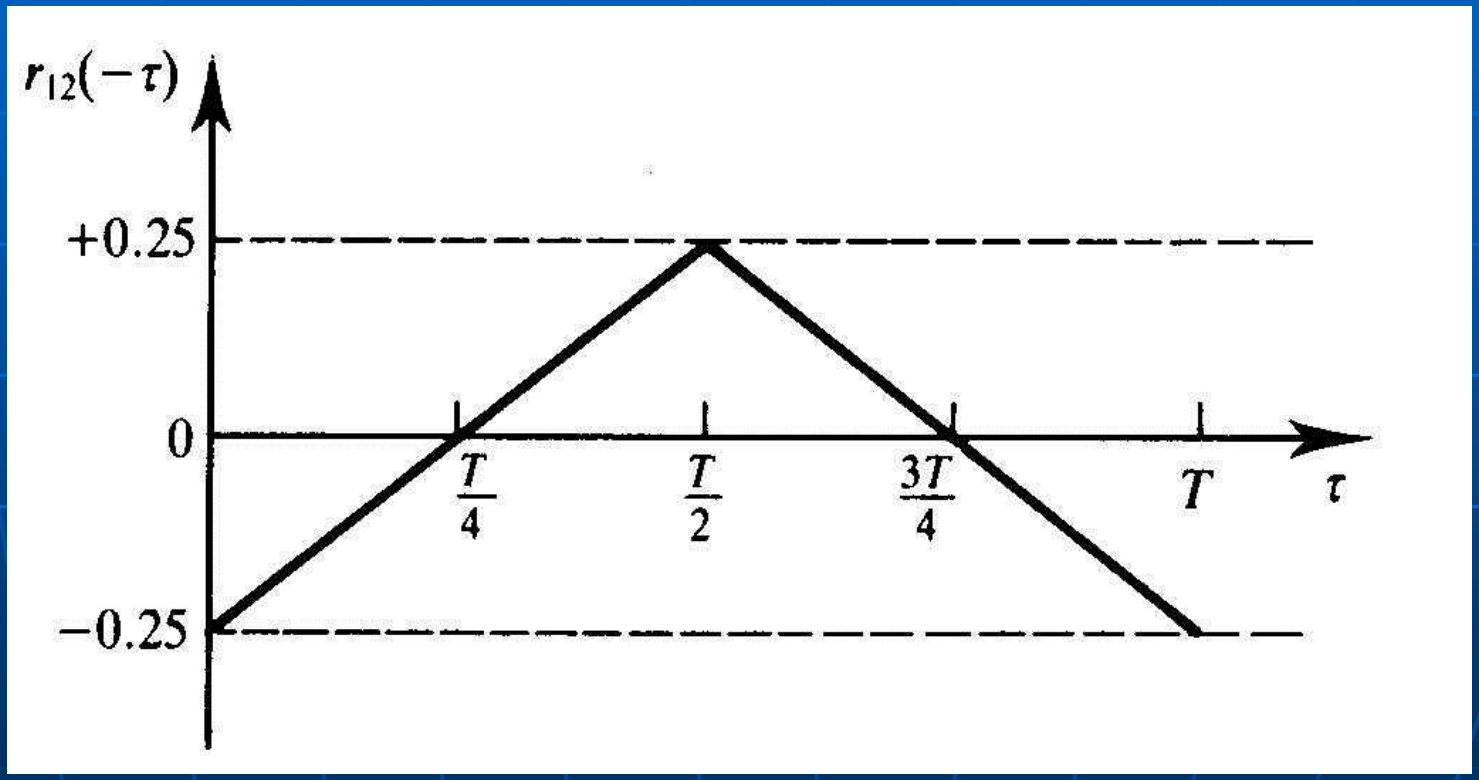


Fig. 3.9 $r_{12}(-\tau)$ as a function of τ .

The cross-correlation coefficient

- The cross-correlation values computed according to the above formulae depend on the absolute values of the data.

- It is often necessary to measure cross-correlations according to the fixed scale between –1 and +1.

- This can be achieved by normalizing the values by an amount depending on the energy content of the data.

Sec. 3.1 Cont. - 36

- **For example,** consider the two pairs of waveforms $x_1(n)$, $x_2(n)$, and $x_3(n)$, $x_4(n)$. The data values are given in the table below:

n	0	1	2	3	4	5	6	7	8
$x_1(n)$	0	3	5	5	5	2	0.5	0.25	0
$x_2(n)$	1	1	1	1	1	0	0	0	1
$x_3(n)$	0	9	15	15	15	6	1.5	0.75	0
$x_4(n)$	2	2	2	2	2	0	0	0	2

Sec. 3.1 Cont. - 37

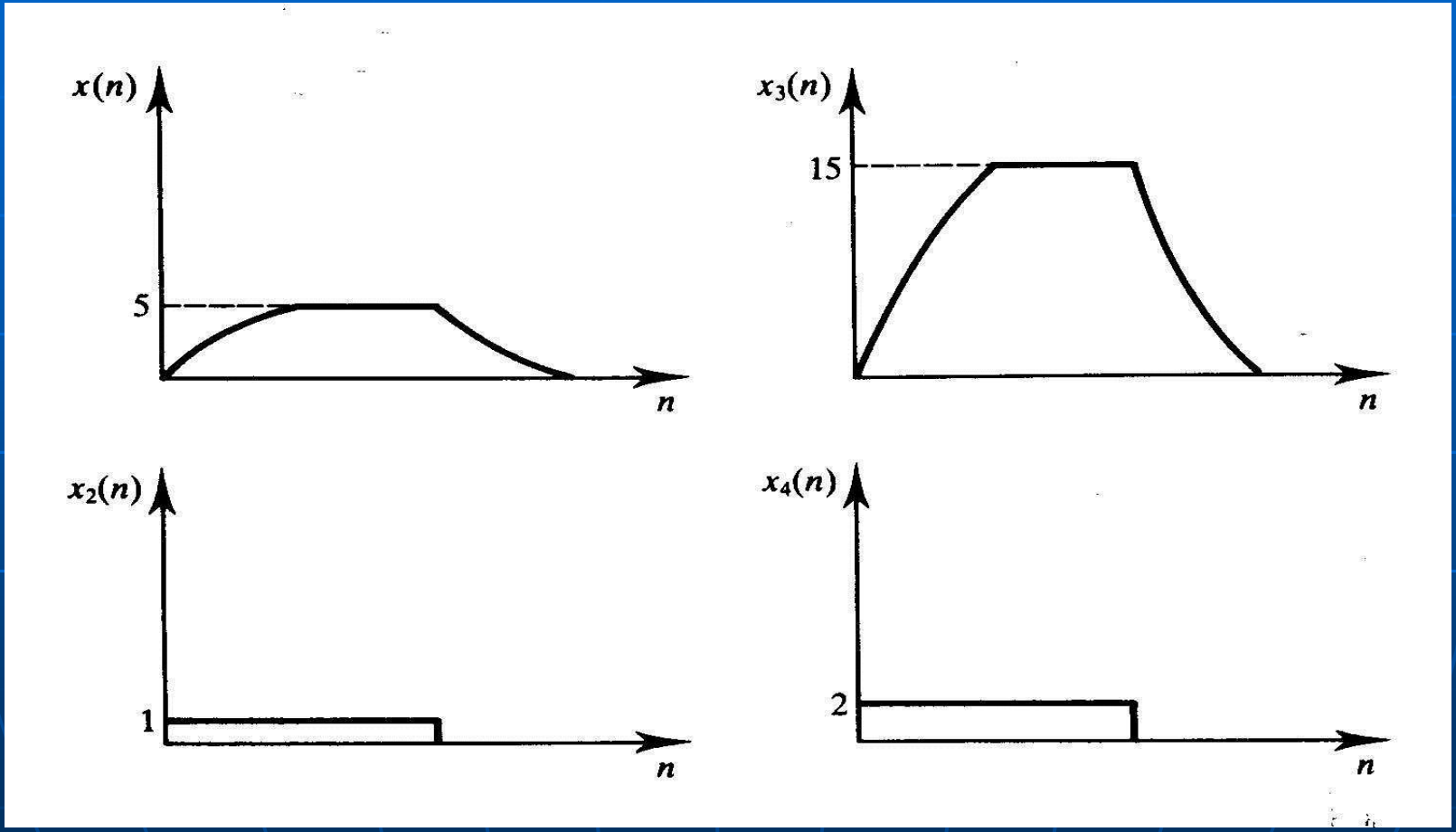


Fig. 3.4 Pairs of waveforms $\{x_1(n), x_2(n)\}$, $\{x_3(n), x_4(n)\}$ of different magnitudes but equal cross-correlations.

Sec. 3.1 Cont. - 38

- As may be seen from Fig. 3.4:

  * waveforms $x_1(n)$ and $x_3(n)$ are alike, differing only in magnitude.

  * The same is true of the pair $x_2(n)$ and $x_4(n)$.

- The correlation, $\mathbf{r_{12}(1)}$ is therefore the same as $\mathbf{r_{34}(1)}$.

Sec. 3.1 Cont. - 39

- However, $r_{12}(1)$ and $r_{34}(1)$ are **1.47** and **8.83** respectively.

- They are different because they depend on the absolute values of the data.

Sec. 3.1 Cont. - 40

- This situation can be rectified by normalizing the cross-correlation $r_{12}(j)$ by the factor:

$$\left[\frac{1}{N}\sum_{n=0}^{N-1} x_1^2(n) \times \frac{1}{N}\sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} = \frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} \qquad (3.9)$$

and similarly for $r_{34}(j)$.

The normalized expression for $r_{12}(j)$ then becomes:

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2}} \qquad (3.10)$$

Sec. 3.1 Cont. - 41

- The normalizing factor for $r_{12}(j)$ in the above illustration is:

$$\frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} = \frac{1}{9}(88.31 \times 6)^{1/2} = 2.56$$

and for $r_{34}(j)$ it is

$$\frac{1}{N}\left[\sum_{n=0}^{N-1} x_3^2(n) \sum_{n=0}^{N-1} x_4^2(n)\right]^{1/2} = \frac{1}{9}(794.8 \times 24)^{1/2} = 15.34$$

Sec. 3.1 Cont. - 42

Therefore

$$\rho_{12}(1) = \frac{r_{12}(1)}{2.56} = \frac{1.47}{2.56} = 0.57$$ **and** $$\rho_{34}(1) = \frac{r_{34}(1)}{15.34} = \frac{8.83}{15.34} = 0.57$$

Now $\rho_{12}(1) = \rho_{34}(1)$ which demonstrates that this normalization process indeed allows a comparison of cross-correlations independently of the absolute data values.

Sec. 3.1 Cont. - 43

$\rho_{12}(j)$ …….. is known as the cross-correlation coefficient. Its value always lies between **–1** and **+1**.

+1 …. means 100% correlation in the same sense,

-1 …. means 100% correlation in the opposing sense, **e. g.** signals in anti-phase.

0…….means zero correlation. This means that the signals are completely independent. This would be the case, **e.g.**, if one of the waveforms were completely random.

Small values of $\rho_{12}(j)$ indicate very low correlation.

The auto-correlation:

- It is a special case occurs when $x_1(n) = x_2(n)$.

- The waveform is then cross-correlated with itself.

- The auto-correlation of a waveform is given by:

$$r_{11}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) x_1(n+j)$$

Sec. 3.1 Cont. - 45

- The autocorrelation function has one very useful property in that

$$r_{11}(0) = \frac{1}{N}\sum_{n=0}^{N-1} x_1^2(n) = S$$

Where
S….. is the normalized energy of the waveform.

- This provides a method for calculating the energy of a signal.

Sec. 3.1 Cont. - 46

- If the waveform is completely random, **e.g.,** corresponding to that of white, Gaussian noise in an electrical system then the autocorrelation will have its peak value at **zero lag** and will reduce to a random fluctuation of small magnitude about zero for lags greater than about unity (**see Fig. 3.5**).

- It is also true that : $r_{11}(0) \geq r_{11}(j)$

Sec. 3.1 Cont. - 47

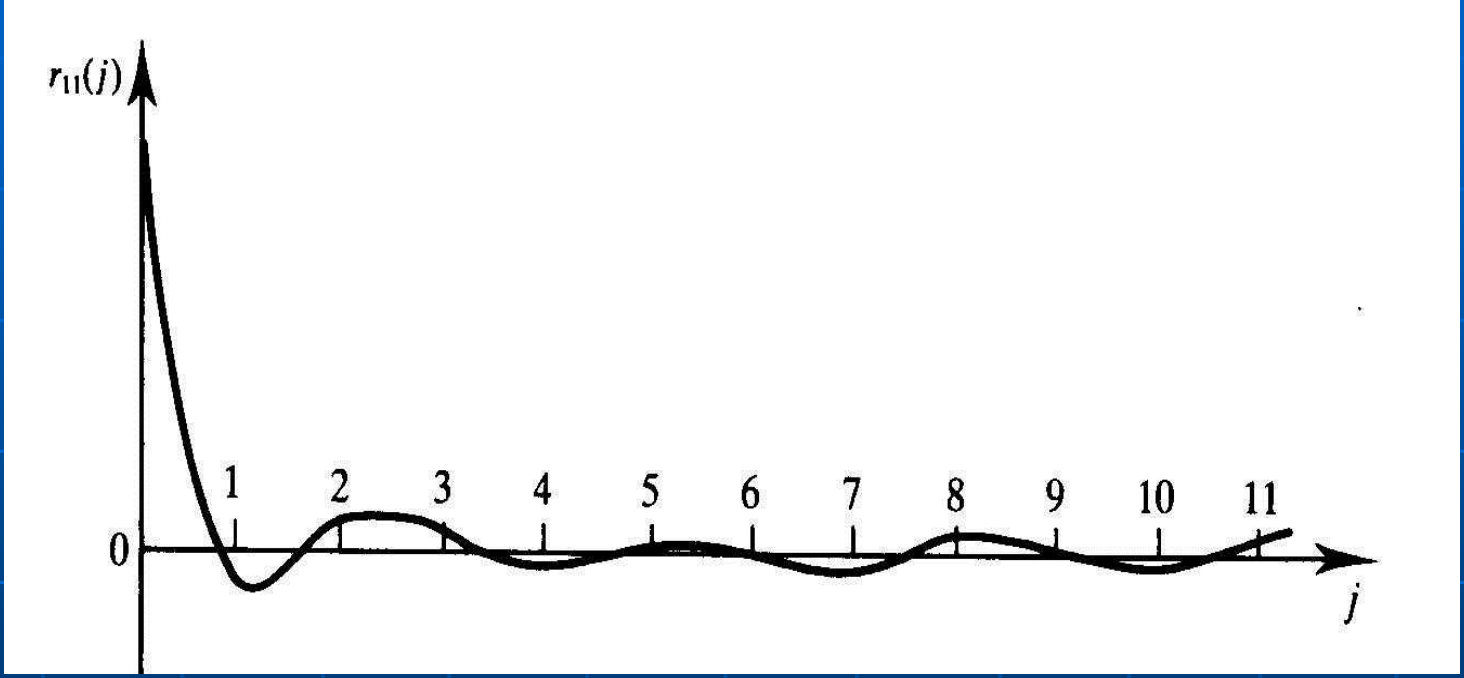


Fig 3.5 Autocorrelation function of a random waveform

3.2 CROSS CORRELATION

- Care has to be exercised when cross-correlating two unequal length sequences when they are periodic.

- This is because the result of the correlation will be cyclic with the period of the shorter sequence.

- This result does not represent the full periodicity of the longer sequence and is, therefore, incorrect.

Sec. 3.2 Cont. - 1

- This may be demonstrated by cross-correlating the sequences a = {4, 3, 1, 6} and **b** = {5, 2, 3} to obtain $r_{ab}(j)$.

- The sequence **b** is placed below sequence a, and **b** is shifted left by one lag on each of the subsequent rows, as shown in the following table.

Sec. 3.2 Cont. - 2

Sequence				Lag	r_{ab} (j)
4	3	1	6		
3	5	2	3	0	47
5	2	3	5	1	59
2	3	5	2	2	34
3	5	2	3	3	47 r_{ab}(j) repeats
5	2	3	5	4	59
etc					

Sec. 3.2 Cont. - 3

- The result shows that $r_{ab}(j)$ is cyclic, repeating every third lag, **i.e.** $r_{ab}(j)$ has the same period as that of the shorter sequence, **b**.

- This procedure is known as **cyclic correlation**.

Sec. 3.2 Cont. - 4

- To obtain the correct value in which each value in **a** is multiplied by each value in **b**, all the elements in **b** have to be shifted in turn below each value in a as shown below:

```
4 3 1 6
      5 2 3
    5 2 3
  5 2 3
5 2 3
  5 2 3
    5 2 3
      5 2 3
```

Sec. 3.2 Cont. - 5

- This is seen to require 6 lags before **"b"** sequence repeats.

- The sequence lengths are 4 and 3 and the number of lags necessary is 4 + 3 – 1 = 6.

- This reveals the general rule for obtaining **the linear cross-correlation** of two periodic sequences of lengths N_1 and N_2 : add augmenting zeros to each sequence to make the lengths of each sequence

$$\mathbf{N_1 + N_2 - 1}.$$

Sec. 3.2 Cont. - 6

- This may be expressed as adding **N_2 – 1 zeros** to the sequence of length **N_1** and

  adding **N_1 – 1 zeros** to the sequence of length **N_2**.

This is now demonstrated for the given sequences **“a”** and **“b”:**

Sec. 3.2 Cont. - 7

- This may be expressed as adding $N_2 - 1$ zeros to the sequence of length N_1 and adding $N_1 - 1$ zeros to the sequence of length N_2.

- As shown in the following table:

Sequence						Lag	R_{ab} (j)
4	3	1	6	0	0		
5	2	3	0	0	0	0	29
2	3	0	0	0	5	1	17
3	0	0	0	5	2	2	12
0	0	0	5	2	3	3	30
0	0	5	2	3	0	4	17
0	5	2	3	0	0	5	35
5	2	3	0	0	0	6	29 r_{ab}(j) repeats
Etc							

Sec. 3.2 Cont. - 8

Sequence						Lag	R_{ab} (j)
4	3	1	6	0	0		
5	2	3	0	0	0	0	29
2	3	0	0	0	5	1	17
3	0	0	0	5	2	2	12
0	0	0	5	2	3	3	30
0	0	5	2	3	0	4	17
0	5	2	3	0	0	5	35
5	2	3	0	0	0	6	29 r_{ab}(j) repeats
Etc							

Thus, the required **linear cross-correlation** of **a** and **b** is :

$$\mathbf{r_{ab}(j)} = \{29, 17, 12, 30, 17, 35\}$$

3.3 FAST CORRELATION

The correlation computation may be speeded up by exploiting the correlation theorem, usually stated as:

$$r_{12}(j) = F_D^{-1}[X_1^*(k)X_2(k)] \qquad (3.13)$$

but which is correctly written as:

$$r_{12}(j) = \frac{1}{N} F_D^{-1}[X_1^*(k)X_2(k)] \qquad (3.14)$$

Sec. 3.3 Cont. - 1

$$r_{12}(j) = \frac{1}{N} F_D^{-1}\left[X_1^*(k)X_2(k)\right] \qquad (3.14)$$

- **Where** F_D^{-1} ..... denotes the inverse DFT.

- This approach requires computation of two DFTs and one inverse DFT, each of which is most easily executed using an FFT algorithm (see Chapter 2).

- If the number of terms in the sequences is sufficiently large, it is quicker to use this FFT method than to calculate the cross-correlation directly.

Sec. 3.3 Cont. - 2

ABSTRACT FOR FAST CORRELATION THEORIM:

- To apply the fast correlation theorem, required:

(1) Calculate: $$X_1^*(k) = \sum_{l=0}^{N-1} x_1(l) e^{j(2\pi/N)lk}$$

(2) Calculate: $$X_2(k) = \sum_{n=0}^{N-1} x_2(n) e^{j(2\pi/N)(-nk)}$$

(3) Calculate: $$X_3(k) = X_1^*(k) X_2(k)$$

(4) Calculate: **X3(n) =** $$F_D^{-1}[X_3(k)]$$

(5) Calculate the correlation as: $$r_{12}(l) = \frac{1}{N} X_3(n)$$

Ch. 3 - 3.3 Cont. - 3

Example 3.4

Find the cross-correlation of the two sequences $x_1(n)$ and $x_2(n)$ below by applying the correlation theorem:

$$x_1(n) = \{1,0,0,1\}$$
$$x_2(n) = \{0.5,1,1,0.5\}$$

Solution

First use the correlation theorem, Eq. 3.12. $X_1(k)$ was found in Section 2.5 to be

$$X_1(k) = (2, 1 + j, 0, 1 - j)$$

and so

$$X_1^*(k) = (2, 1 - j, 0, 1 + j)$$

Ch. 3 - 3.3 Cont. - 4

Example 3.4 - Cont - 1

- $X_2(k)$ is obtained using the FFT algorithm given in section 2.5. $X_2(k) = (3, -0.5 - j0.5, 0, -0.5 + j0.5)$

- Bringing the values of the FFTs together gives

$$X_1^*(k) = (2, 1 - j, 0, 1 + j)$$

$$X_2(k) = (3, -0.5 - j0.5, 0, -0.5 + j0.5)$$

Thus

$$[X_1^*(k)\ X_2(k)] = (6, -1, 0, -1)$$

Ch. 3 - 3.3 Cont. - 5

Example 3.4 - Cont - 2

The components of $F_D^{-1}\left[X_1^*(k)X_2(k)\right]$

are obtained by dividing the obtained values by **N = 4**.

Thus $F_D^{-1}\left[X_1^*(k)X_2(k)\right] = (1,\ 1.5,\ 2,\ 1.5)$

Now , from Equation 3.12,

$$r_{12}(j) = \frac{1}{4} F_D^{-1}\left[X_1^*(k)X_2(k)\right] = \{0.25,\ 0.375,\ 0.5,\ 0.375\} \qquad (3.26)$$

Example 3.4 - Cont - 3

$$r_{12}(j) = \frac{1}{4} F_D^{-1}\left[X_1^*(k) X_1(k)\right] = \{0.25, 0.375, 0.5, 0.375\} \qquad (3.26)$$

- This correlation will be **circular** because all the data is periodic with period N.

- The cross-correlation $r_{12}(j)$ may be worked out directly to be

$$r_{12}(0) = (1 \times 0.5 + 0 + 0 + 1 \times 0.5)/4 = 0.25$$

$$r_{12}(1) = (1 \times 1 + 0 + 0 + 1 \times 0.5)/4 = 0.375$$

$$r_{12}(2) = (1 \times 1 + 0 + 0 + 1 \times 1)/4 = 0.5$$

$$r_{12}(3) = (1 \times 0.5 + 0 + 0 + 1 \times 1)/4 = 0.375$$

Example 3.4 - Cont - 4

- The next value, $r_{12}(4)$, is **0.25**, the same as $r_{12}(0)$, and the sequence repeats periodically.

- This is **circular correlation**, as discussed in Section 3.2, and this result agrees with that derived above using the **correlation theorem.**

Example 3.4 - Cont - 5

- The **correlation theorem** can be used to obtain the **linear correlation** by adding augmenting zeros to the two sequences as explained in Section 3.2.

- Thus, if the sequence lengths are N1 for $x_1(n)$ and N_2 of $x_2(n)$, then $N_2 - 1$ zeros are added to $x_1(n)$ and N1 – 1 zeros are added to $x_2(n)$.

- The cross-correlation is then computed using these two augmented sequences.

- This method of evaluating cross-correlations by using the correlation theorem and FFTs is known as **fast correlation.**